

# SPARE PART CATALOG

16 V 4000 S83L

527110199E



*Power. Passion. Partnership.*

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

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## Preface

This catalog is the basic reference document for ordering spare parts. Illustrations are provided for extensive assembly operations to assist trained personal in conducting such operations but do not claim to be complete. For better overview, even product components that are not available as spare parts are shown.

The individual components were taken from the assembly list.

## Catalog structure

The catalog is divided into Function Groups (FG) and Function Group Segments (FGS) in ascending order.

The side contain the following information:

- **MODEL:** (Product Type)
- **SERIAL NO.:**
- **FG/FGS DESIGNATION**
- **FG/FGS NO.:**
- **BOM No.:**

## Explanation of columns

- **ITEM:** Provides a cross-reference between illustration and text section.
- **PART NO.:** This is the valid number for ordering the part in question.
- **DESIGNATION:** The precise part name.
- **MODEL/TYPE** is an additional specification.
- **QTY.: (Quantity in Unit):** Indicates the quantity of this part required for the case in question. Quantities not stated are available on request.
- **UNIT:** The 3-character codes indicate: PC = Piece, KG= Kilogram, M= Meter, ROL= Roll, PAC= Pack
- **WEIGHT**
- **FN.:** (Footnote) Provides additional parts information. For clarification of footnote symbols, see Page 2.
- **DIMENSIONS:** This column contains standard-part dimensions or manufacturer designations for purchased parts.
- **STANDARD**

## Amendments service

Part No. changes and amendments which occur after publication of the catalog are not published. We therefore recommend that users make a record of amendments in their catalogs. This information is provided, for instance, in order confirmations and/or on spare parts labels.

**Footnotes: Key**

|            |                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AB</b>  | Only available complete. Repair by manufacturer only.                                                                                                                                                                                                                                                          |
| <b>AC</b>  | Not scheduled as a spare part. Availability must be checked.                                                                                                                                                                                                                                                   |
| <b>ACN</b> | Not scheduled as a spare part. Availability must be checked. Component only available complete.                                                                                                                                                                                                                |
| <b>EM</b>  | Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file.<br><br>When ordering, quote EM code with order no. E.g: 535 030 04 17 EM. |
| <b>EMN</b> | Emission-relevant parts. Component only available complete.                                                                                                                                                                                                                                                    |
| <b>KD</b>  | Group balancing required                                                                                                                                                                                                                                                                                       |
| <b>KE</b>  | Individual part; not scheduled as a spare part.                                                                                                                                                                                                                                                                |
| <b>KF</b>  | Adjusted spare part.                                                                                                                                                                                                                                                                                           |
| <b>KFN</b> | Adjusted spare part. Component only available complete.                                                                                                                                                                                                                                                        |
| <b>M</b>   | Spares-related components or parts with different repair sizes.<br>Dimensions are listed in the Tolerances and Wear Limits List.                                                                                                                                                                               |
| <b>MN</b>  | Spares-related components or parts with different repair sizes. Components only available complete.                                                                                                                                                                                                            |
| <b>MUB</b> | Spares-related components or Parts with different repair sizes. Available only as complete set.                                                                                                                                                                                                                |
| <b>NN</b>  | Component is not broken down into further sub-components.                                                                                                                                                                                                                                                      |
| <b>P</b>   | List of bought-in components                                                                                                                                                                                                                                                                                   |
| <b>SB</b>  | Group of parts which are welded or brazed together                                                                                                                                                                                                                                                             |
| <b>UA</b>  | Indicate dimensions when ordering                                                                                                                                                                                                                                                                              |
| <b>UB</b>  | Only supplied complete or as a complete set.                                                                                                                                                                                                                                                                   |

## Ordering Procedure

The following information is essential for spare-part order processing:

- Engine Model
- Serial number (see engine name plate)
- FG/FGS
- Item
- Part No.
- Designation
- Model/Type
- Quantity
- Page

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

**When ordering a calibrated part** (marked by "KF", "KFN" in the "FN" column), type the following additional information in the **Comments** field:

If the calibration No. is known (see part's nameplate):

Add the calibration No., preceded by an oblique stroke, as an extension to the part number ("/xxxxx")

e.g.: 0025362810/E1234 Here, the extension "E1234" would be the part's calibration number on the nameplate.

If the calibration No. is not known:

Give a detailed description of the required part's installation location by quoting the complete Device Description code (e.g. "= 37.1.1 + A022 - F003") plus the part number extension "/77" (e.g. 5005362810/77).

In this case, the calibrated spare part will be supplied under a new part number.

The Device Description is found on the nameplate at the part's installation location, or in the corresponding Wiring Diagram.

When ordering a non-calibrated adjustable part (marked by "KF", "KFN" in the "FN" column) which should not be calibrated by the factory: Add extension "/76" to the part number (e.g. 0025362810/76).

Phone orders should be confirmed in writing to prevent errors.

## Ordering Example:

| Engine model | Serial number   | FG/FGS | ITEM | Part No.   | Designation       | Quantity | Page |
|--------------|-----------------|--------|------|------------|-------------------|----------|------|
| 16V4000R41   | 527101836       | 031 88 | 5    | 5270300701 | Crankshaft        | 1        | 33   |
|              |                 | 083 05 | 10   | 0020922801 | Easy-Cange Filter | 2        | 92   |
| Blue Line    | V4000M Waterjet | 517 00 | 75   | 5185301593 | Control Console   | 1        | 53   |

## Factory repairs

Parts shipped to MTU for repair must be marked with the relevant engine serial number (tag or label, for example).

### **Exchange components (Reman)**

Exchange components are parts overhauled by the manufacturer. They have the same service life and quality characteristics as new parts.

Components which are included in the Reman exchange program are marked by the following letters in front of the item number:

For instance:       **E 524 010 3520 = New component in exchange program**

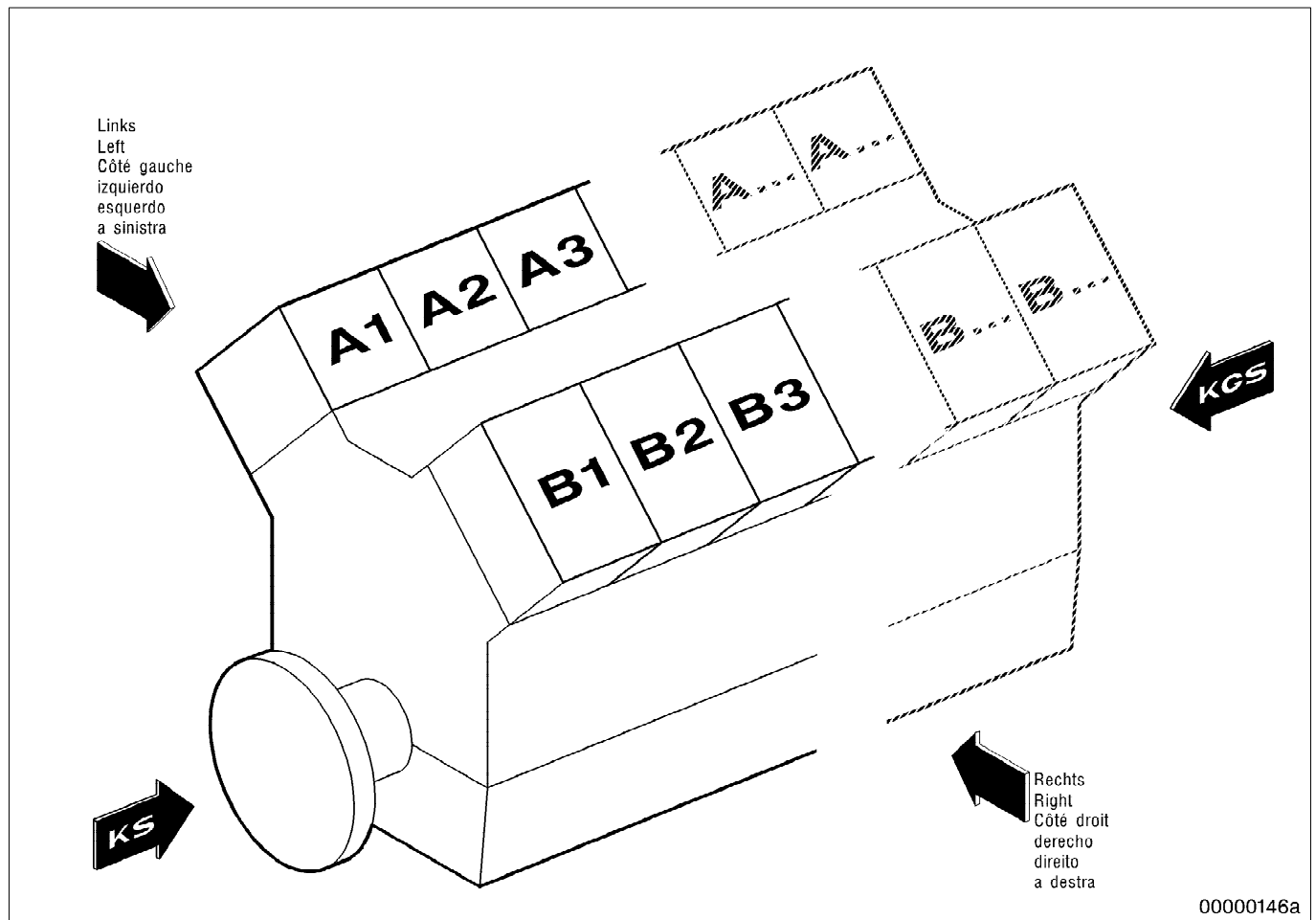
**R 524 010 3520 = Reman component in exchange program**

### **Exchange procedure**

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

Further information upon request.

## Engine side and cylinder designations



Engine sides / banks are always designated as viewed from the driving end (KS).

The left bank of cylinders is designated "A" and the right bank "B" (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively from the driving end, starting with No. 1.

Other engine components are also numbered consecutively from the driving end, starting with No. 1.

### Designations and abbreviations

Driving end = KS

Free end = KGS

Left-hand side

Right-hand side

Top

Bottom

Turbocharger = ATL



|     |                                                             |
|-----|-------------------------------------------------------------|
| FG  | FG DESIGNATION                                              |
| 004 | LABELS AND AUXILIARY MATERIALS                              |
| 010 | CRANKCASE AND ATTACHMENTS                                   |
| 011 | CRANKCASE                                                   |
| 014 | OIL PAN                                                     |
| 015 | ENGINE MOUNTING                                             |
| 016 | END HOUSING, DRIVING END AND FREE END                       |
| 018 | CRANKCASE BREATHER                                          |
| 024 | GEARCASE, AUX PTO END                                       |
| 031 | CRANKSHAFT WITH BEARINGS                                    |
| 032 | CRANKSHAFT, PTO ON DRIVING END                              |
| 035 | CRANKSHAFT, PTO ON AUX PTO END                              |
| 036 | TRANSPORT LOCK/LOCK FOR CRANKSHAFT                          |
| 037 | PISTON                                                      |
| 038 | CONNECTING ROD                                              |
| 041 | CYLINDER HEAD                                               |
| 051 | CAMSHAFT                                                    |
| 055 | VALVE DRIVE                                                 |
| 070 | FUEL SYSTEM (HIGH PRESSURE)                                 |
| 073 | INJECTION PUMP / HP PUMP                                    |
| 075 | FUEL INJECTOR                                               |
| 080 | L.P. FUEL SYSTEM; (ALSO GAS ENGINES)                        |
| 081 | FUEL PIPEWORK BEFORE DELIVERY PUMP                          |
| 082 | FUEL PIPEWORK FROM DELIVERY PUMP TO FILTER                  |
| 083 | FUEL FILTER                                                 |
| 087 | LEAK-OFF FUEL MONITORING                                    |
| 100 | TURBOCHARGING (EXHAUST TURBOCHARGER / COMPRESSOR)           |
| 111 | CHARGE AIR COOLING, GENERAL, LEFT SIDE                      |
| 121 | AIR FILTER                                                  |
| 124 | AIR SUPPLY TO CYLINDERS                                     |
| 125 | AIR ROUTING BETWEEN TURBOCHARGER AND CHARGE-AIR COOLER      |
| 141 | EXHAUST SYSTEM AFTER CYLINDER HEAD                          |
| 142 | CONSTANT-PRESSURE EXHAUST MANIFOLD                          |
| 172 | ELECTRIC STARTING                                           |
| 181 | LUBE OIL PUMP AND DRIVE                                     |
| 183 | OIL FILTRATION/COOLING                                      |
| 184 | CRANKCASE-INTERNAL OIL SUPPLY                               |
| 185 | TURBOCHARGER OIL SUPPLY                                     |
| 202 | COOLANT CIRCUIT, GENERAL, HT CIRCUIT                        |
| 203 | LOW-TEMPERATURE CIRCUIT                                     |
| 206 | COOLANT PIPEWORK WITH THERMOSTAT                            |
| 213 | GENERATOR                                                   |
| 231 | ENGINE MOUNTING                                             |
| 240 | ENGINE/GEARBOX/ENGINE/GENERATOR CONNECTION                  |
| 364 | FUEL SUPPLY SYSTEM (FOR PERIPHERAL APPLICATIONS)            |
| 500 | MONITORING AND CONTROL SYSTEM, GENERAL ELECTRICAL EQUIPMENT |
| 507 | WIRING, GENERAL, FOR ENGINE/GEARBOX/PLANT                   |

**FG FG DESIGNATION**

508 ACCESSORIES FOR ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM

511 ALARM SYSTEM